

emgforge: an open, validated forward model of surface EMG — from volume-conductor lead fields to motor-unit action potentials and interference signals

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Abstract

We describe **emgforge**, an open-source forward model that turns tissue geometry into surface and intramuscular electromyograms: a volume conductor (an analytical multilayer cylinder or a finite-element model of a parametric limb or a segmented MRI forearm) gives, by reciprocity, the lead field $\varphi(z)$ each muscle fibre sees from an electrode; a line-source model of the fibre turns φ into a single-fibre action potential (SFAP); fibres are grouped into motor units, motor units are driven by a phenomenological motoneuron pool with a twitch model, and the result is a motor-unit action potential (MUAP) library, high-density EMG on an electrode grid, force, and static or non-stationary interference EMG. The centre of the work is one synthesis recipe — the spatial line-source integral with a monopole-fitted lead field, a second-derivative current source, a one-sided tendon window and physical time — which we justify step by step and then test. Against a closed-form line-source oracle the recipe reproduces the extracellular potential with $r = 1.0000$ and zero lag, its source is monopole-free to machine precision, its end-of-fibre potential starts at L/v within 0.4 ms, and the conduction velocity, innervation-zone signature, end-of-fibre behaviour, depth decay, fat attenuation, electrode-size and inter-electrode-distance laws all agree with the quantitative EMG literature. We also report, plainly, what the tests found wrong: the amplitude constant of the recipe is off by exactly $1/v$; the Fourier-domain engine we inherited from the literature is anti-phase and lagged with respect to first principles; on finite-element lead fields the recipe's amplitude is erratic at the $\pm 40\%$ level even though its shape and spectrum are faithful; and the forearm MUAPs, though now of physiological duration, are still several times smaller than recorded ones. Three datasets are released with the code: a cylinder SFAP atlas, a 100-motor-unit forearm pool with its 5×5 HD-EMG MUAP tensor, and interference-EMG trials. The report is meant to be read with the repository and updated as the open items close.

1 Introduction

Forward models of the electromyogram simulate the signal an electrode records when muscle fibres fire. They are used to understand what shapes a motor-unit action potential, to test decomposition and estimation algorithms on signals whose ground truth is known, and, increasingly, to generate labelled training data for learning-based EMG decoding [1–3]. The classical models are analytical: a fibre in an infinite anisotropic medium [4–6], a planar layered conductor [7, 8], or a multilayer cylinder [9–11]. Finite-element models lift the geometric restrictions [12–14] and, with reciprocity and reusable source bases, can now be run on MRI-derived anatomy at

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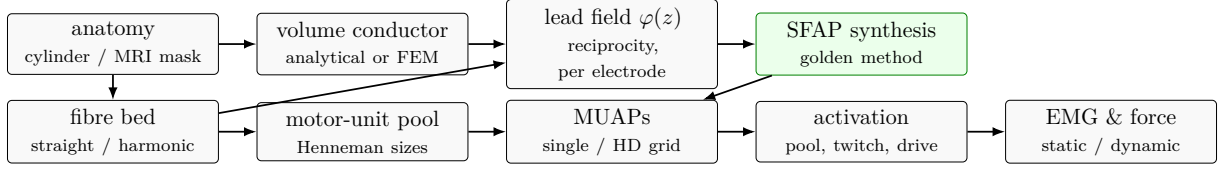


Figure 1: The `emgforge` chain. The lead field is the hinge: it is computed once per electrode and sampled along every fibre; the golden synthesis recipe turns each fibre’s $\varphi(z)$ into an SFAP; fibres are summed per motor unit, and the activation layer turns motor-unit spike trains into interference EMG and force.

interactive speed [1]. Generative surrogates trained on such twins [2, 3] make the output cheap, but they inherit whatever the twin got right or wrong, and the twin is usually closed.

`emgforge` is our attempt at an open, inspectable version of that chain. Its design rule is that every step must be checkable in isolation: the lead field against an analytical model, the synthesis integral against a closed-form solution, the MUAP features against numbers from the physiological literature, and the whole chain against the statistics of real interference EMG. This report is written to accompany the public repository while a fuller paper is prepared. It is deliberately plain: it says what the model computes, in what order, with which parameters, and what happened when we tested it.

The contributions are:

1. a complete, documented forward chain (Sec. 2) with three interchangeable volume conductors (analytical cylinder, parametric FEM, MRI forearm FEM) behind one lead-field interface;
2. a single SFAP synthesis recipe — the *golden method* — derived from the line-source model and justified step by step (Sec. 4);
3. a fibre-geometry layer that includes anatomically contained harmonic-streamline fibres from a segmentation mask (Sec. 5);
4. a phenomenological activation layer that gives spike trains, force and static or non-stationary interference EMG (Sec. 6);
5. a tiered validation suite with literature-anchored numeric criteria and a first-principles oracle, run end to end and reported honestly (Sec. 7);
6. three released datasets (Sec. 8).

2 Overview of the pipeline

Figure 1 shows the chain. Everything downstream of the volume conductor is pure NumPy; only the finite-element solve needs FEniCSx [15] and Gmsh [16].

Reciprocity. By the reciprocity theorem for a linear volume conductor [17], the potential at electrode e due to a unit current source at a fibre point $\mathbf{r}$ equals the potential at $\mathbf{r}$ due to a unit current injected at e . So one solve with the source at the electrode gives the *lead field* $\varphi_e(\mathbf{r})$ everywhere, and sampling it along a fibre path $\mathbf{r}(z)$ gives $\varphi_e(z)$, the weight with which membrane current at z reaches the electrode. Every fibre, every motor unit and every contraction reuse the same solve; only the number of electrodes costs finite-element work. The interior-point reciprocity of our solver is checked directly (Sec. 7.1).

Table 1: Volume-conductor parameters. The validation cylinder uses the analytical values in both models; the production FEM/MRI table follows Gabriel et al. [20] at ≈ 100 Hz.

	bone	muscle $\sigma_{\perp}$	muscle $\sigma_{\parallel}$	fat	skin	radii (mm)
analytical [11]	0.02	0.10	0.50	0.04	1.0	10 / 35 / 38 / 40
FEM, validation cylinder	0.02	0.10	0.50	0.04	1.0	9, 10 / 35 / 38 / 40; $L = 240$
FEM / MRI, production	0.02 [†]	0.2455	1.228	0.0379	4.55×10^{-4}	from the segmentation

[†]cortical; cancellous bone 0.075 S/m.

3 Volume conductors and lead fields

3.1 Analytical multilayer cylinder

The reference conductor is the four-layer cylinder of Farina et al. [11] (bone, anisotropic muscle, fat, skin), solved in the spatial-frequency domain with Bessel functions; **emgforge** carries a line-by-line Python port of the original implementation (**emgforge.analytical**). We use it in two ways: its own signal generator produces MUAPs directly (the “Farina generator”), and its transfer function can be inverted to give the lead field $\varphi(z)$ along a fibre for a given electrode, which we feed to our own synthesis engine. The geometry and conductivities are in Table 1. The electrode is a disc of radius 5 mm; the coupled grid $\Delta z = v/f_s = 0.977$ mm (with $v = 4$ m/s, $f_s = 4096$ Hz) and 256 samples give a 250 mm span along the fibre.

3.2 Finite-element conductors

The FEM tier solves the quasi-static conductivity equation

$$\nabla \cdot (\boldsymbol{\sigma}(\mathbf{r}) \nabla u) = -s(\mathbf{r}), \quad (\boldsymbol{\sigma} \nabla u) \cdot \mathbf{n} = 0 \text{ on } \partial\Omega, \quad (1)$$

on a tetrahedral mesh with piecewise-constant tissue tensors (muscle transversely isotropic, ratio 5 along the fibre [18, 19]; fat, skin, bone isotropic). The source s is a zero-mean Gaussian blob of width σ_s centred on the electrode (its mean is subtracted so the pure-Neumann problem is solvable), and the linear system is solved with GMRES/ILU in FEniCSx [15]. Two families of mesh are built with Gmsh [16]: a *parametric limb* — layered discs (cancellous and cortical bone, muscle, fat, skin) extruded or lofted along z , circular or elliptical, optionally tapered, refined at the skin — and the *MRI forearm* (Sec. 3.3). For the cylinder used in validation the radii and conductivities are set equal to the analytical model’s (Table 1); the mesh has 1.03×10^6 cells and a solve takes ≈ 12 s. Figure 2 shows the resulting lead fields against the analytical ones. The width of the electrode source matters more than one might expect: with $\sigma_s = 5$ mm the blob extends through the 2 mm skin and 3 mm fat into the muscle, and its transverse footprint at the fibre depth is narrower than the analytical one; a $\sigma_s = 1$ mm source reproduces the analytical transverse profile (Fig. 2c).

3.3 MRI forearm

A labelled segmentation (muscles, radius and ulna, fat/skin) is meshed into a tetrahedral forearm with a thin skin shell; each muscle’s conductivity tensor is rotated onto that muscle’s own centreline tangent, estimated from per-slice centroids and a smoothing spline (**sigma_mode="centerline"**). Skin electrodes are placed on the outer surface by ray-casting onto the mesh; the HD-EMG layout used in this report is a regular 5×5 grid with 10 mm inter-electrode distance centred over the flexor carpi ulnaris (FCU), rows along the arm and columns around it (measured spacing 10.2 mm along, 10.0 mm across). The forearm mesh used here (subject WR, 42,315 cells) builds in 5 s and solves in 0.6 s per electrode; sampling φ along

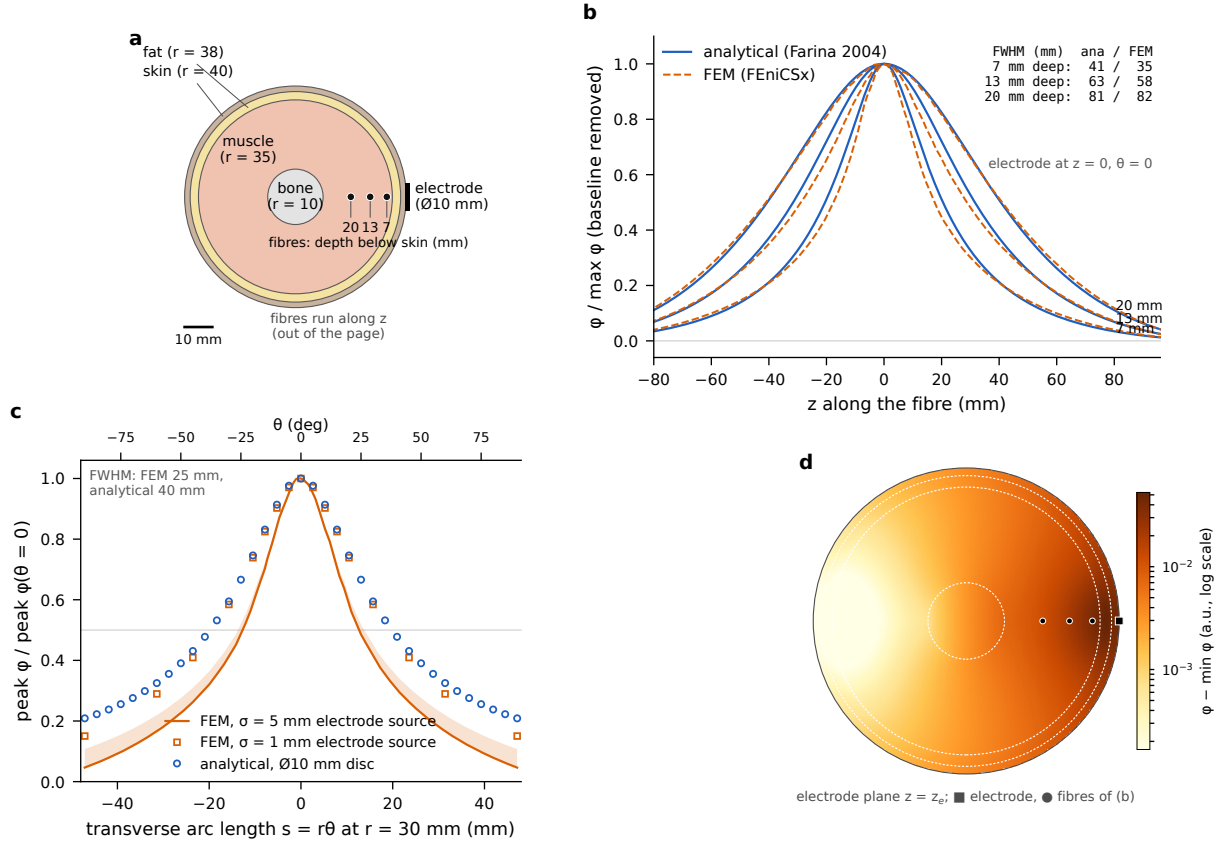


Figure 2: Lead fields on the validation cylinder. (a) Cross-section with the electrode and the fibre depths used. (b) $\phi(z)$ along fibres 7, 13 and 20 mm below the skin: analytical (solid) and FEM (dashed); the widths agree for deep fibres and differ for the shallowest, where the electrode models differ (a $\varnothing 10$ mm disc against a Gaussian source that behaves as a $\varnothing 5$ mm disc). (c) Transverse decay at the fibre depth ($r = 30$ mm): the FEM field around the circumference (FWHM 25 mm; the band is the baseline uncertainty) against the analytical peak potential for fibres displaced around the circumference (FWHM 40 mm); with a $\sigma_s = 1$ mm source the FEM profile matches the analytical one (extra points), so the difference is the 5 mm electrode blob, not the conductor. (d) The FEM lead field of one electrode on the cross-section (log scale); the dots are the fibres of (b).

the 637 fibres of the FCU bed takes 6 s per electrode. A fresh solve of this model reproduces the lead fields distributed with the NeuroDec digital twin [1] for the same subject with $|r| = 1.000$ over 105 electrodes $\times$ 100 fibres (Sec. 7.4).

4 The golden SFAP synthesis method

4.1 Line-source model

A fibre of radius a and intracellular conductivity σ_{in} carrying a transmembrane potential $V_{\text{m}}(z, t)$ injects, per unit length, the membrane current of core-conductor theory [4, 21]

$$i_{\text{m}}(z, t) = \sigma_{\text{in}} \pi a^2 \frac{\partial^2 V_{\text{m}}}{\partial z^2}, \quad (2)$$

and the potential at the electrode is the lead-field-weighted sum of that current:

$$\text{SFAP}(t) = \int i_{\text{m}}(z, t) \varphi(z) \, dz = \sigma_{\text{in}} \pi a^2 \int V_{\text{m}}(z, t) w(z) \varphi''(z) \, dz, \quad (3)$$

where the second form follows by integrating by parts twice (the boundary terms vanish because the windowed potential has compact support) and shows that the SFAP is the inner product of the action potential with the *second derivative* of the lead field. The action potential is the Rosenfalck form $V_{\text{m}}(s) = A s^3 e^{-s}$ for $s > 0$ (s in mm, $A = 96$ mV) launched from the neuromuscular junction at z_0 in both directions,

$$V_{\text{m}}(z, t) = V_{\text{m}}(vt - |z - z_0|) w(z), \quad w(z) = \text{window over } [z_0 - L_1, z_0 + L_2], \quad (4)$$

so that $t = 0$ is the instant the junction fires, the wave front reaches the proximal tendon at L_1/v and the distal one at L_2/v . Taking $\partial^2/\partial z^2$ of the whole windowed field, rather than of each travelling half analytically, is what produces the three source terms that the literature separates by hand [22, 23]: the two propagating tripoles, the *generation* term at the cusp $|z - z_0|$, and the *end-of-fibre* terms where the window cuts the wave. These are exactly the terms that make the source monopole-free, $\int i_{\text{m}} \, dz = 0$ for all t [24]; we verify this to 8×10^{-17} (Sec. 7.1).

4.2 The recipe

Discretised, Eq. (3) is one matrix–vector product, $\text{SFAP} = (\mathbf{C} \boldsymbol{\varphi}) \Delta z$, with $\mathbf{C}$ the $(n_t \times n_z)$ current-source-density matrix. The steps around it, and the reason for each, are (Fig. 3, Table 2):

1. **Monopole denoise of φ .** A finite-element $\varphi(z)$ carries mesh-scale ripple. Because the SFAP integrates φ'' , that ripple is amplified quadratically in spatial frequency. Instead of low-passing — which rounds the real peak and, as we show in Sec. 7.2, halves the spectral content — we replace φ by a least-squares fit of the form the field actually has, $\hat{\varphi}(z) = \sum_{i=1}^3 A_i / \sqrt{d_i^2 + (z - z_i)^2} + c$, a sum of three free-position point sources. On a clean analytical φ the fit is a near no-op ($r > 0.99$); on a FEM φ it changes the field itself by at most 1.8% of its peak but lifts the correlation of φ'' with the analytical one from 0.47 to 0.91, cuts the ripple in φ'' eighteen-fold, and takes the SFAP’s jaggedness from 0.011 to 0.002 (the analytical value is 0.0018).
2. **Edge taper.** A short cosine ramp (5 and 10 samples) on the two ends of the sampled φ , a Gibbs guard for lead fields truncated by the sampling window.
3. **Upsampling $\times 2$ (cubic spline).** The second derivative needs a finer grid than the raw φ or the SFAP develops a Nyquist zigzag; $\times 2$ removes it and $\times 4$ adds nothing.

Table 2: The golden recipe (`SpatialConfig`). The cylinder tier uses $f_s = 4096$ Hz; the MRI tier 2048 Hz with $v = 3.3\text{--}4$ m/s.

parameter	value	why
<code>denoise</code>	monopole, 3 poles	fit the analytic form of φ ; removes mesh ripple before φ''
<code>edge_taper_left/right</code>	5 / 10 samples	Gibbs guard at the sampling-window ends
<code>upsample_factor</code>	2	second derivative off the Nyquist zigzag
<code>csd_derivative</code>	2	physical current source (triphasic)
<code>fiber_window</code>	one-sided, $\alpha = 0.25$	flat at the junction, tapered at the tendons
<code>center_time / t_start_ms</code>	false / -10	physical time, junction at $t = 0$
v, f_s, w	4 m/s, 4096 Hz, 256	coupled grid $\Delta z = v/f_s$
σ_{in}, a	1, 0.05 mm	amplitude constant (see Sec. 9)

4. **Current-source density.** $\mathbf{C}$ is built from the full bidirectional windowed field of Eq. (4) by two numerical z -derivatives.
5. **One-sided tendon window.** The window $w(z)$ is flat through the junction and cosine-tapered over the last 25% of each semi-fibre only. A symmetric Tukey window would notch the junction, where the wave is born; a hard boxcar produces a sharp end-of-fibre spike (physically right for one fibre with a sharp tendon, and the choice we use when comparing to sharp-tendon oracles) which in real muscle is softened by tendon scatter. The taper costs $r = 0.998$ against the sharp-tendon oracle.
6. **Physical time.** No centring: $t = 0$ is the junction, the detector lobe of an electrode Δz from the junction lands at $\Delta z/v$, the end-of-fibre onset at L/v ; a 10 ms pre-roll is prepended so a junction under the electrode is not pinned to the window edge.

4.3 What the recipe is not: the Fourier engine

`emgforge` also carries the spatial-frequency-domain engine of Farina and Merletti [8], Farina et al. [11], in which φ is transformed along z , multiplied by the fibre-end function, the action-potential spectrum and jk_z , and brought back to time by a Radon section. It was our first production engine and the reference the spatial engine was originally tuned against. The validation in Sec. 7.1 shows that, given the same φ , it does not reproduce the line-source integral: it is anti-phase and lagged by 1–2.5 ms, its end-of-fibre onset leads L/v by 2.5 ms, and its propagating main lobe has the opposite sign to the textbook convention. We therefore treat the spatial recipe as the reference and keep the Fourier engine only for comparison.

4.4 From SFAPs to a MUAP

A *fibre bed* object carries, per fibre, the sampling step, the two semi-lengths, the junction position and the conduction velocity; a MUAP is the sum of the SFAPs of the fibres in a motor unit, each computed from its own $\varphi(z)$ with the shared configuration. Junction scatter, conduction-velocity scatter and tendon scatter are therefore explicit per-fibre values rather than filters: a 5 mm junction scatter across 50 fibres lengthens the MUAP from 20.0 to 22.2 ms and reduces its peak-to-peak to 0.86 of the coherent sum (Sec. 7.2), the temporal dispersion real MUAPs show. Because time is physical, this dispersion is real and not a window artefact — the reason the spatial engine, and not the Fourier one, carries production.

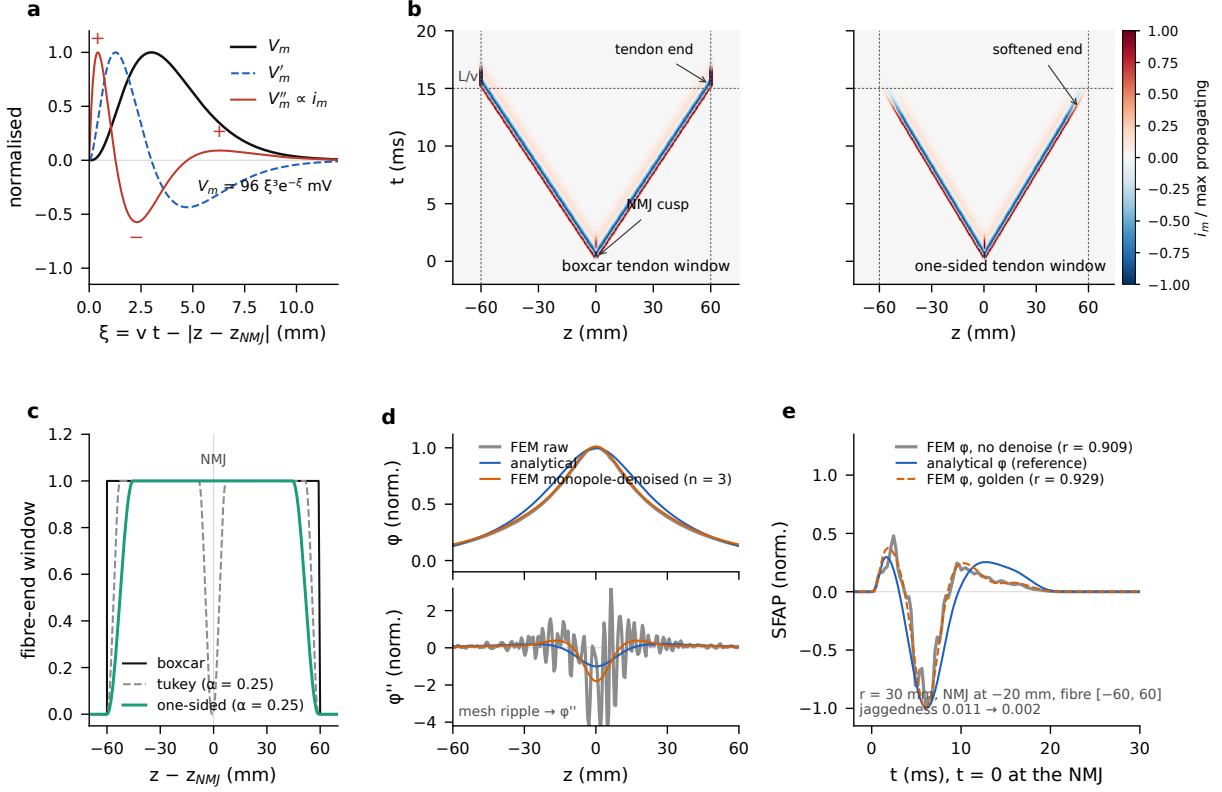


Figure 3: Anatomy of the golden method. (a) The Rosenfalck action potential and its first and second spatial derivatives; the second derivative is the triphasic current source. (b) The current-source-density matrix $C(z, t)$ for a fibre $[-60, 60]$ mm with the junction at 0: two tripoles leave the junction, the cusp at the junction and the window edges at the tendons appear as sources; boxcar (left) and one-sided (right) tendon windows. (c) The three fibre-end windows. (d) A finite-element lead field raw, monopole-denoised and analytical, with their second derivatives. (e) The SFAPs each produces: the ripple in ϕ'' reaches the raw SFAP and the monopole fit removes it.

5 Fibre geometry and motor units

5.1 Straight fibre beds

For a parametric limb, and as the default for the MRI forearm, fibres are placed on the muscle’s cross-section at mid-length by Poisson-disk sampling at a chosen density (4 fibres/mm² for the FCU, which oversamples the pool: the 100-unit pool only needs a subset) and extended as full-length paths parallel to the muscle centreline, with one global semi-length and a junction at an atlas innervation-zone fraction (0.305 of the length for the FCU). This is the classical assumption; it is simple, and it is what the literature’s cylinder models assume. Its cost is discussed in Sec. 9.

5.2 Harmonic-streamline fibres

Straight, constant-depth, full-length fibres are a poor description of a real muscle belly. Noura [surname] contributed a method that resolves a muscle’s internal architecture from its label mask alone: solve the masked Laplace problem $\nabla^2 \phi = 0$ inside the muscle with $\phi = 0$ on the proximal cap, $\phi = 1$ on the distal cap and an insulated surface, and trace the streamlines of $\nabla \phi$ (second-order Runge–Kutta) from seeds on the mid-belly cross-section. Streamlines are contained in the muscle and never cross by construction. In the production model each streamline is one full-length fibre with a single junction placed at the muscle’s shared innervation zone; a quality guard rejects tracer failures (turn per step above 15°, or a longitudinal span below 30% of the muscle). An experimental variant cuts each streamline into short in-series fibres, each with its own junction at an atlas innervation band — the series-fiberling that long muscles show [25]; it narrows the surface field to real motor-unit widths but is not yet validated, and is gated behind a flag. On the FCU, seeded densely enough to match the straight bed (0.5 mm seed grid; the library default of 2 mm gives only 52 streamlines), the harmonic bed has 827 fibres that stay inside the mask for 99.96% of their length against 96.6% for the 637 straight fibres; their junctions lie within 1.2 mm (s.d.) of the shared innervation zone. With the junction at that zone the two beds give near-identical MUAPs ($\approx 0.26 \mu\text{V}$ for a matched unit), so the harmonic advantage at this stage is anatomical rather than electrical (Fig. 4). Note that the production harmonic model places the junction at mid-length by default while the straight bed uses the atlas fraction (0.305); the two are made consistent by passing the same fraction, and the released datasets use the straight bed.

5.3 Motor-unit pool

Motor units are drawn from the bed following the size principle [26]: N sizes log-uniform between 5 and 400 fibres (many small units, few large), each unit anchored on a random bed fibre and its territory radius grown until it contains the target number of fibres, territories allowed to overlap; units are sorted by size so that index i is recruitment order. A unit’s MUAP is the sum of its fibres’ SFAPs; Fig. 5 shows the pool and its MUAPs, and the amplitude–size relation.

6 Activation and interference EMG

The activation layer is phenomenological, written from the equations of the Fuglevand pool as parametrised in NeuroMotion [3, 26]. Given a scalar excitation $E(t) \in [0, 1]$:

- *Recruitment*: unit i has a threshold RTE_i spread exponentially with range 50 up to $E = 0.75$ for the last unit.
- *Rate coding*: once recruited, the rate is $\min(\text{PFR}_i, \text{MFR}_i + (E - \text{RTE}_i) s_i)$ with the peak rate falling from 40 to 30 pps, the minimum from 10 to 5 pps and the slope from 50 to 30 pps per unit drive across the pool — the onion skin [27].

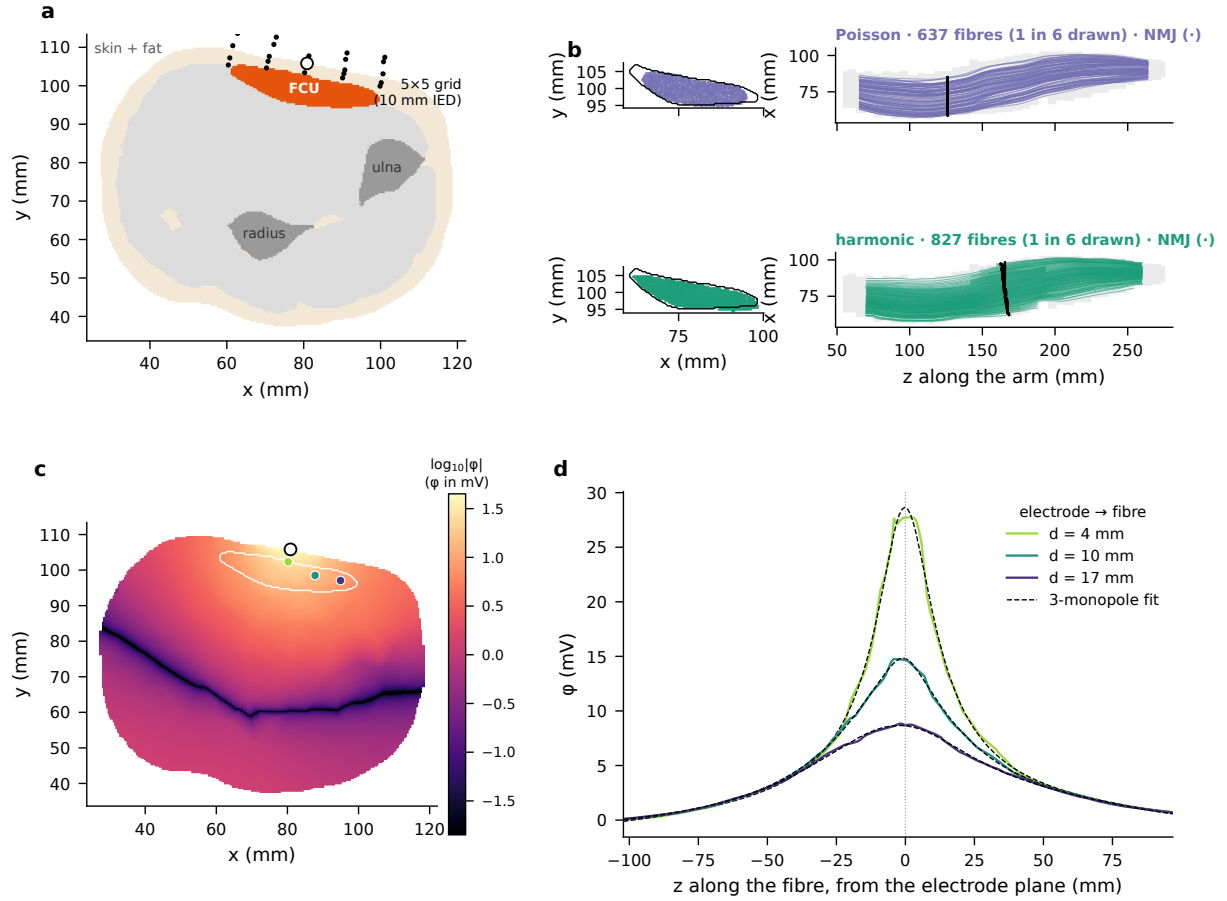


Figure 4: The MRI forearm (subject WR). (a) A segmentation slice with the FCU highlighted and the 5×5 electrode grid projected. (b) The FCU fibre bed at 4 fibres/mm²: 637 straight Poisson fibres and 827 harmonic streamlines (one in six drawn), with the junctions marked — at the atlas fraction 0.305 of the length for the straight bed and, by default, at mid-length for the harmonic one (the atlas value can be passed). (c) The reciprocal lead field of one grid electrode on a mesh slice (log scale). (d) ϕ along three fibres 4, 10 and 17 mm from the electrode, with the three-monopole fit the golden recipe uses (dashed).

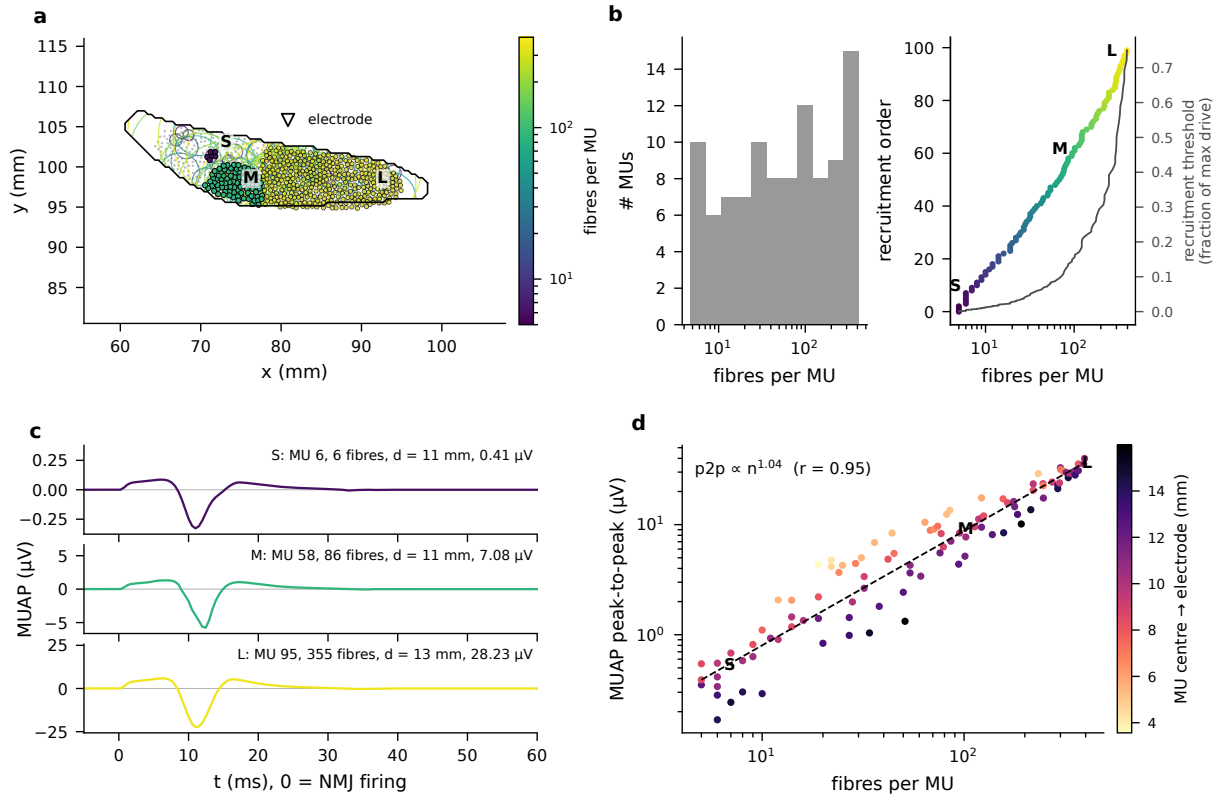


Figure 5: The 100-unit FCU pool. (a) Territories on the cross-section, colour by size (5–395 fibres, median 66; territory radii 0.6–13.8 mm), with the centre electrode of the grid. (b) Size distribution and recruitment order against size. (c) Golden MUAPs at the centre electrode of a small, a medium and a large unit (0.4, 7 and 28 μ V; across the pool 0.17–40 μ V, median 5.9, median duration 18 ms). (d) MUAP amplitude against fibre count: $p2p \propto n^{1.04}$ ($r = 0.95$), the scatter being depth (colour).

- *Spikes*: a renewal process with Gaussian inter-spike intervals of coefficient of variation $1/6$, re-drawn at each discharge; units de-recruit when E drops below threshold. A common-drive fluctuation (low-passed noise at 2 Hz) can be added to E .
- *Force*: each discharge adds a twitch $P_i(t/T_i) e^{1-t/T_i}$ with amplitudes spanning $100\times$ and larger units contracting faster, scaled by the rate-dependent fusion gain of Fuglevand et al. [26]; the sum is normalised to the mean force at full excitation, so force is in %MVC.
- *EMG*: each spike places the unit's MUAP (single channel or on every electrode of the grid) and the trains sum; single- and double-differential montages are differences of electrodes.
- *Non-stationary EMG*: a normalised joint angle $\theta(t)$ sets the drive $(0.08+0.42\theta)$, scales each spike's MUAP $(1 + 0.7(\theta - 0.5))$; shortening brings fibres to the electrode) and time-warps it $(1 + 0.18(\theta - 0.5))$. This is a modulation layer, not a musculoskeletal model.

Figure 6 shows a unit on the grid, an interference plateau, the pool's recruitment and rate coding, a trapezoid contraction with its force, and a movement trial.

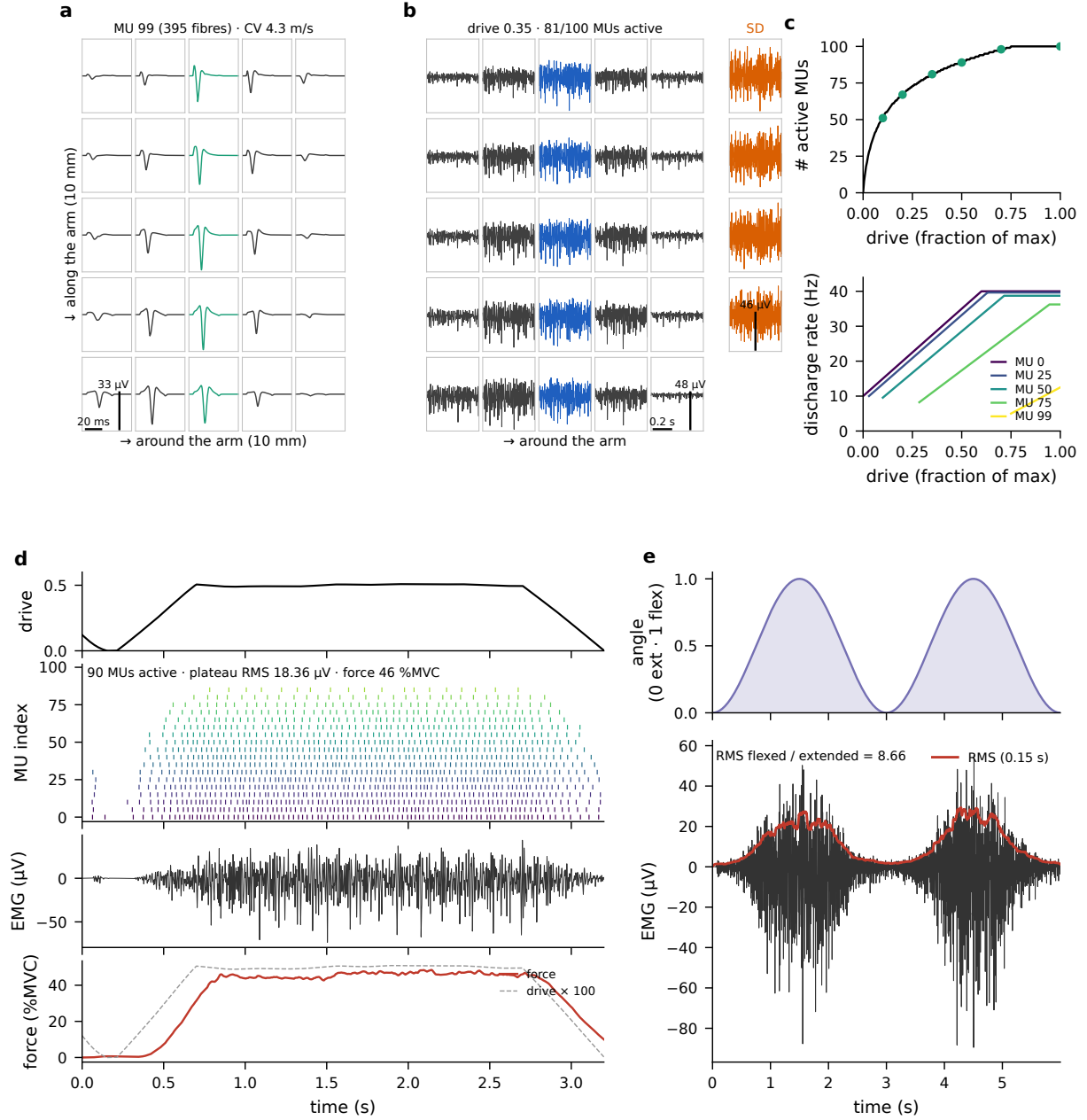


Figure 6: From motor units to signals (regular 5×5 grid, 10 mm spacing). (a) The largest unit's MUAP on the grid: rows run along the arm, so the wave walks down a column (conduction velocity read off the grid: 4.3 m/s for this unit, 4.1–4.9 over the pool); columns run around it, so the footprint decays sideways. (b) Interference HD-EMG at a 35%-drive plateau (81 of 100 units active), and a single-differential column. (c) Recruitment and rate coding of the pool. (d) A trapezoid contraction at half drive: drive, spikes, EMG (plateau RMS 18 μ V), force (46% MVC); across drives 0.1–1.0 the plateau RMS is 3.4–33 μ V and the force 7–100% MVC. (e) A movement trial: angle and non-stationary EMG (RMS flexed/extended 8.7).

7 Validation

The suite (`scripts/validation/`, ≈ 4 min) holds one measurable property of the chain per check against a literature principle with a numeric criterion, in four tiers: (A) the cylinder must reproduce — first principles, then the analytical model, then the finite-element model, then the pipeline on each; (B) the MUAP phenomenology the literature agrees on; (C) the statistics of interference EMG and the pool; (S) the chain-level sanity checks that ship with the simulator. Table 3 is the scoreboard; the full report with every measured number is in the repository (`docs/validation/`). A “known” flag marks a documented limitation that keeps its criterion, so it turns green when fixed, without failing the run.

7.1 Tier A: the cylinder must reproduce

First principles. A fibre in an infinite anisotropic medium has a closed-form lead field, $\varphi(z) \propto (\rho^2/\sigma_\rho + z^2/\sigma_z)^{-1/2}$ [19, 21], whose second derivative is analytic. Evaluating Eq. (3) in the dual form on a 0.02 mm grid gives an oracle with no numerical derivative of the tendon step and no volume-conductor model in the loop. The spatial engine, given the same φ on its 0.977 mm grid, matches it with $r = 1.0000, 0.9999, 0.9999$ and lags 0.00, -0.05 , -0.05 ms for junctions 0, -20 , -30 mm from the electrode (Fig. 7a). The source is monopole-free to 7.7×10^{-17} ; the SFAP is invariant to the sampling rate and step ($r > 0.999$, amplitude within 0.2%); moving the electrode by 20 mm delays the propagating lobe by 4.88 ms ($\Delta z/v = 5.00$, one sample); superposition is exact; and changing v from 4 to 3 m/s stretches the SFAP by 1.328 (4/3) and scales its mean frequency by 0.757 (3/4) [28]. The one thing the engine gets wrong is its constant: the ratio to the oracle is 0.497 at $v = 2$ and 0.249 at $v = 4$ — exactly $1/v$ (Fig. 7c; Sec. 9). The Fourier engine on the same φ scores $r = -0.79, -0.68, -0.64$ at lags of 1.4, 2.6, 0.9 ms (Fig. 7b).

Analytical against finite-element lead fields. Along fibres 7–20 mm below the skin the FEM $\varphi(z)$ correlates with the analytical one at $r = 0.995$ – 1.000 , and at $r = 0.997$ – 1.000 for fibres 10 – 45° off the electrode meridian. Their widths differ for the shallowest fibres (FWHM 41 vs 35 mm at 7 mm), and comparing the second derivatives against analytical electrodes of different radii shows the FEM Gaussian source behaves as a 5 mm-diameter disc; at mid-depth (13 mm) φ'' agrees only to $r = 0.93$ whatever electrode is assumed — the one tier-A failure. The pipeline run on FEM φ against the pipeline run on analytical φ gives $r = 0.90$ – 0.97 for shallow fibres and 0.998 for deep ones; the Fourier route, which low-passes φ at ≈ 32 mm wavelength, gives 0.995– 1.000 at all depths, which is how earlier reports of “ $r \approx 0.99$ ” were obtained.

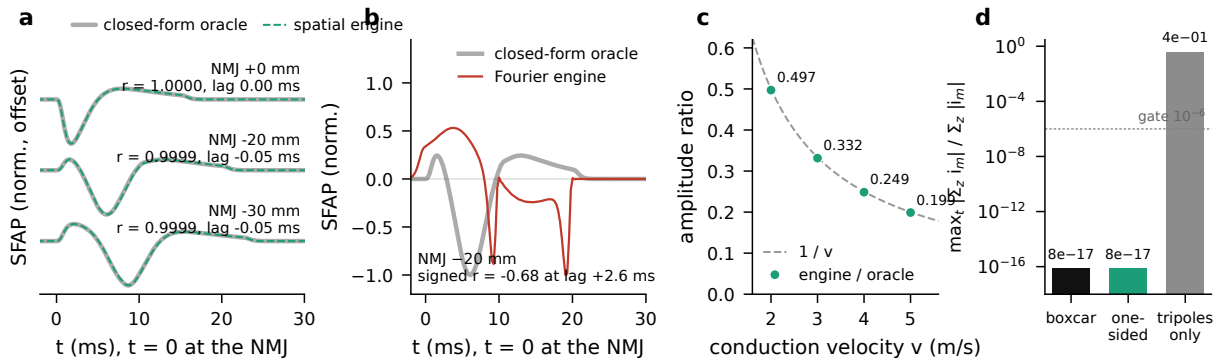


Figure 7: First principles. (a) The closed-form line-source SFAP and the spatial engine, three junction positions. (b) The Fourier engine against the same oracle. (c) The engine-to-oracle amplitude ratio follows $1/v$ (ratio $\times v = 0.994$ at every v). (d) The net source current at every instant is zero to machine precision once the junction and end terms are included; the propagating tripoles alone leave 39%.

Table 3: Validation scoreboard (branch `paper/arxiv-v1`; tiers C and S run on the released forearm pool, dataset D2). The two genuine failures are discussed in Sec. 9.

tier	scope	pass / gated	known	fails
A	cylinder: first principles, analytical, FEM, pipeline	14 / 15	4	A1.1b: FEM φ'' at mid-depth, $r = 0.93$
B	MUAP features vs literature	14 / 14	0	—
C	interference EMG and pool	4 / 5	1	C2: no refractory floor
S	chain-level sanity	7 / 7	1	—

Timing and symmetries. Isolating the end-of-fibre component by subtracting the same fibre with the tendon moved 60 mm further, its onset lands at 10.26 and 15.39 ms for $L/v = 10$ and 15 in the pipeline; in the Farina generator it leads by up to 2.5 ms. Conduction velocity from a four-electrode array is 3.94 m/s (pipeline) and 3.83 m/s (Farina) for $v = 4$. Rotating the electrode by 30° equals rotating the fibre ($r = 1.000$, peak within 1%); shifting it 20 mm along the axis equals shifting the sampling window ($r = 0.996$ – 0.999 , the residual being the 240 mm mesh’s end effect); and the lead field between two interior points is reciprocal to 1.35%.

7.2 Tier B: MUAP features against the literature

Each check is run on the Farina generator (the field’s reference model) and on our pipeline, with arrays and montages built by translating the lead field (Fig. 9, Fig. 8):

- *Morphology*: between the junction and the tendon the monopolar SFAP is $+ - +$ with the main lobe negative [29]. The pipeline is; the Farina generator’s main propagating lobe is positive.
- *Conduction velocity* from a single-differential array recovers 3.13, 3.98, 4.88 m/s for set values 3, 4, 5 [30].
- *Innervation zone*: monopolar potentials mirror about the junction ($r > 0.998$); the single-differential channel over it is 4% of the maximum and the channels either side are mirror images of opposite sign ($r < -0.998$) [31].
- *End of fibre*: the same onset on every electrode (0.00 ms spread); its ratio to the propagating part grows monotonically with depth (0.02 at 7 mm to 0.20 at 20 mm) and is suppressed mono $>$ SD $>$ DD (0.053, 0.014, 0.004) [32, 33].
- *Depth*: peak-to-peak amplitude is a power law in depth over 7–25 mm ($R^2 > 0.98$), exponent 2.9 monopolar and 3.5 single-differential [25, 34].
- *Fat*: RMS retained at 3/9/18 mm of fat is 0.50/0.23/0.12 against the finite-element curve 0.69/0.20/0.10 of Kuiken et al. [35]; mean frequency falls; the FEM tier attenuates like the analytical one within 30% once φ is smoothed.
- *Electrode and montage*: amplitude and mean frequency fall with electrode radius; a 5 mm disc is -2.8 dB at 100 cycles/m [29]; single-differential amplitude doubles from IED 2.5 to 5 mm and saturates beyond 10 mm; with a single travelling wave, the two electrodes of a bipolar pair see shift-copies to 1.4×10^{-3} , so the montage is the textbook comb filter with nulls at nv/IED [28, 36].
- *Anisotropy* elongates the lead field along the fibres by exactly $\sqrt{5}$ in the infinite medium and by 1.5 in the layered cylinder.
- *Linearity*: 50 identical fibres give $50\times$ the SFAP to 10^{-15} .

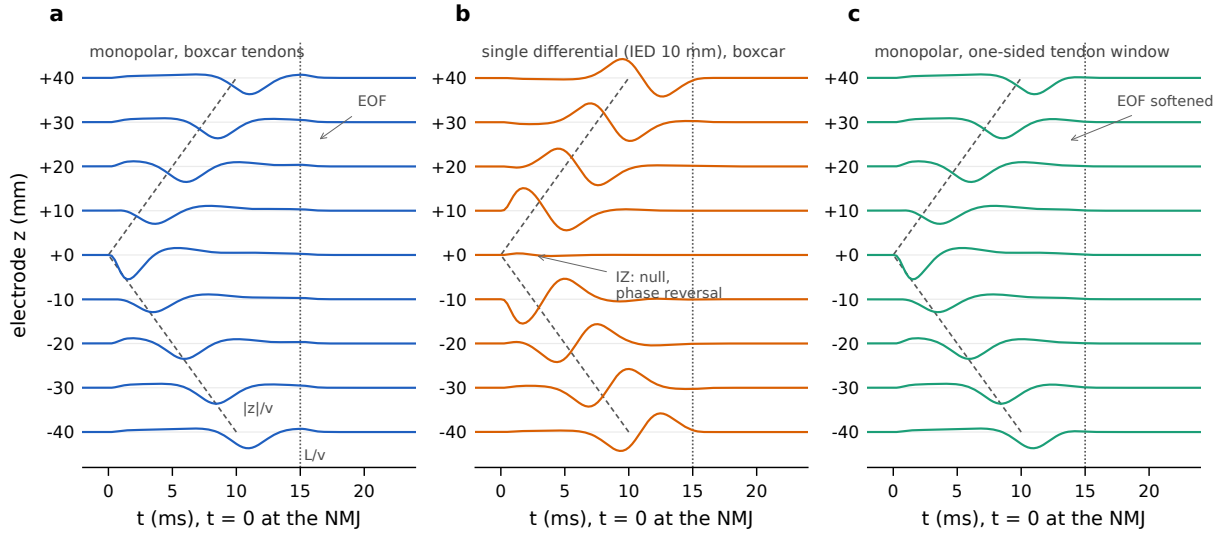


Figure 8: Physical time on a finite-element lead field (fibre 10 mm deep, tendons at ± 60 mm, $v = 4$ m/s). (a) Monopolar SFAPs along a nine-electrode array with hard tendons: the propagating lobe walks at $|z|/v$ (a fit gives 4.04 m/s) while the end-of-fibre potential starts at 15.4–15.6 ms on every channel ($L/v = 15$ ms; spread 0.24 ms). (b) The single-differential montage vanishes and reverses at the innervation zone. (c) The same array with the one-sided tendon window of the golden recipe: the end-of-fibre potential falls from 0.16 to 0.10 of the propagating amplitude.

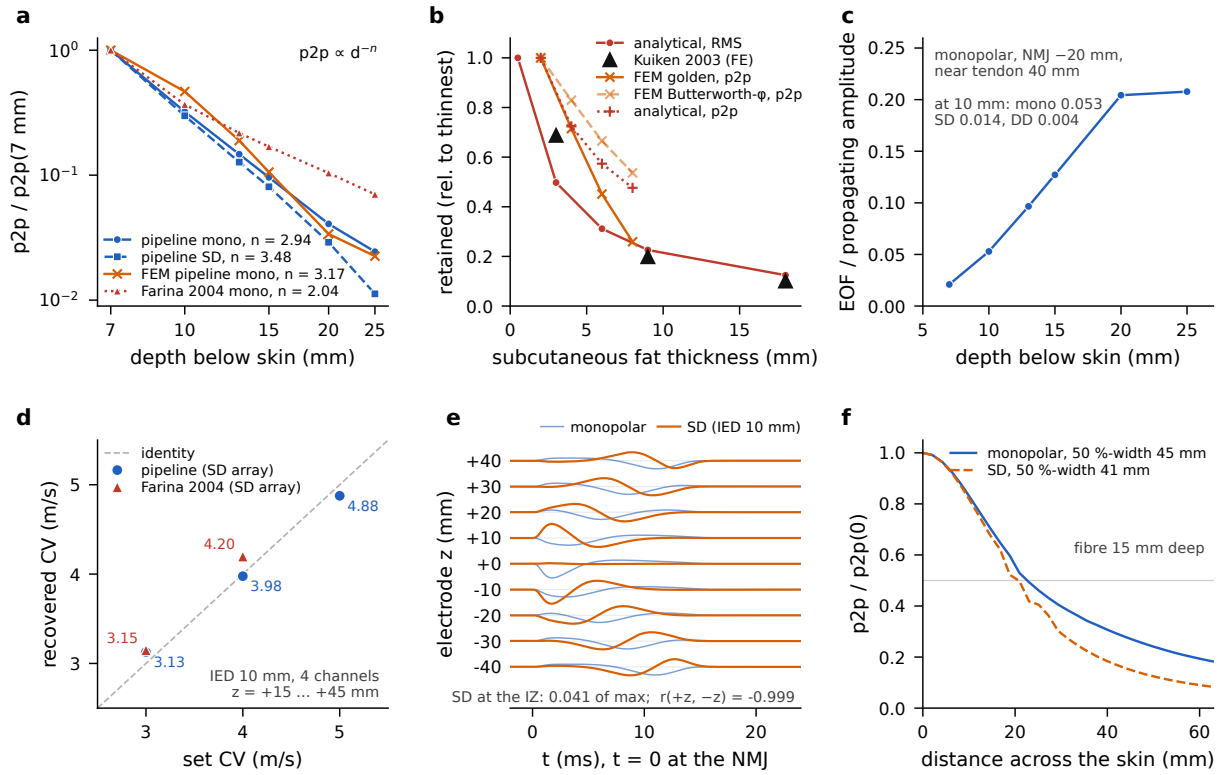


Figure 9: MUAP features against the literature. Depth law (log-log), fat attenuation against Kuiken et al. [35], end-of-fibre ratio against depth, conduction velocity recovered against set, the single-differential array across the innervation zone, and the transverse spread of monopolar and single-differential signals.

7.3 Tier C and S: interference EMG and the chain

These tiers run on the released forearm pool (Sec. 8). The pool’s rates lie in 5–40 pps with the onion-skin ordering ($\rho = -0.98$ between threshold and rate); inter-spike-interval variability is 0.16; thresholds are right-skewed and twitches span $95\times$; the amplitude distribution is super-Gaussian at low drive (kurtosis 5.6 at 5%) and Gaussian at high (2.9–3.1 at 50–90%), with ARV/RMS 0.77–0.79, between the Laplacian and Gaussian values [37, 38]; the RMS at half maximal force is 0.56 of that at maximal force, inside the linear-to-quadratic band of Lawrence and De Luca [39]; the median frequency at moderate drive is 77 Hz (mean 81 Hz); larger units give larger MUAPs ($\rho = 0.98$). The chain-level checks pass: orderly recruitment, onion skin, EMG rising with drive, force monotone and calibrated to 100% MVC at full drive, conduction velocity 4.33 m/s read off the HD grid (4.1–4.9 over all 100 units, 10 mm inter-electrode distance), a localised MUAP footprint (peak/mean RMS 2.2), and an $8.7\times$ amplitude modulation across a movement. Amplitude cancellation is flagged (68% already at 20% drive against $\approx 33\%$ in Keenan et al. [40]); see Sec. 9.

7.4 MRI tier

A fresh solve of the WR forearm model reproduces the lead fields distributed with the NeuroDec digital twin [1] for the same subject at $|r| = 1.000$ for all 105 electrodes and 100 fibres — the volume conductor is the same physics on the same anatomy. On the same anatomy the harmonic fibres are contained in the muscle mask for 99.9% of their length (96.7% for straight fibres), and the mesh-resolution study of the earlier pipeline gave $r = 0.9998$ between 10^5 and 2×10^6 -cell meshes.

8 Released datasets

Three datasets are generated by scripts in `paper/figures/` and described by manifests in `paper/datasets/` (Table 4). They are meant as reference material: every array can be re-generated from the repository.

9 Limitations and open items

We list these in the order we would fix them.

1. **Amplitude constant.** The engine computes $(\mathbf{C}\varphi)\Delta z/v$, dividing by v once too often: $\mathbf{C}$ is already the second spatial derivative, so for a spatially fixed action potential the potential does not depend on v . The ratio to first principles is exactly $1/v$. A physiological dependence on conduction velocity [6] should come from stretching the action potential in space with v , not from a prefactor. Absolute amplitudes in this report should be read with that factor in mind.
2. **Fourier engine.** Given a lead field, it is anti-phase and lagged relative to the line-source integral, and its end-of-fibre onset is early. We suspect the orientation and sign conventions of the action potential in the Radon formulation (`V2 = -flip(...)`); until audited it should not be used as an oracle.
3. **FEM lead-field fidelity for the recipe.** On finite-element φ the golden SFAP amplitude is erratic at $\pm 40\%$ across depth: the recipe integrates φ'' faithfully down to ≈ 8 mm wavelengths, where FEM φ carries mesh structure that the three-pole fit does not remove (five or seven poles do not either). Low-passing φ makes amplitudes smooth but halves the SFAP’s mean frequency (46 vs 112 Hz golden, 91 Hz analytical). A finer mesh around the fibre band, or a smoothing-free local estimate of φ'' , is the fix. Until then FEM amplitude

Table 4: Datasets released with the report (float32, NumPy .npz with a JSON manifest each; 8 MB in total).

name	contents	size
cylinder_sfap_atlas	analytical and FEM lead fields and golden SFAPs on the validation cylinder for 6 depths $\times$ 5 angles $\times$ 3 junction offsets $\times$ 9 electrode positions (1620 SFAPs), with the Farina-generator MUAP of each case	1.6 MB
forearm_fcu_mu_pool	the 100-unit FCU pool (fibre paths, fibre indices, sizes, territories, activation parameters), golden MUAPs at the centre electrode and the full (100, 25, 256) HD-grid MUAP tensor, electrode positions, configuration	2.4 MB
interference_emg_trials	trapezoid contractions at six drive levels (0.1–1.0): drive, spike trains, single-channel and 25-channel EMG, force; one movement trial	4.1 MB

laws should be read through the smoothed route; shape, timing and spectrum can be read from the golden one.

4. **Depth law.** The FEM cylinder decays $\approx 30\%$ slower with depth than the analytical one over 7–20 mm, whatever the preprocessing or the source width. This is a volume-conductor difference still to be attributed (mesh resolution near the skin against the analytical model’s spectral truncation). The source width does explain a different discrepancy — the transverse footprint (Fig. 2c) — so the cylinder tier should move to a 1 mm source, or a true surface patch.
5. **MUAP scale and fibre geometry.** With the grid centred on the muscle, the forearm pool’s MUAPs span 0.17–40 μV (median 5.9) and last 18 ms; real surface MUAPs are tens to hundreds of microvolts (the largest 1–2 mV) and 5–20 ms. The duration is now in range; the amplitude is several times low, of which the $1/v$ constant (item 1) accounts for a factor ≈ 4 and the rest is the straight, full-length, constant-depth fibre geometry that the series-fiber experiment addresses. Amplitude cancellation inherits it (68% at 20% drive, where Keenan et al. [40] give 33%): 100 units firing 18 ms MUAPs overlap more than a real pool’s. We note for the record that an earlier grid layout in the repository (electrodes snapped to mesh vertices, its centre 17 mm off the muscle) gave 0.6 μV and 33 ms — an electrode-placement artefact that was first read as a model property; the sanity suite’s “C-04” flag is now the amplitude only.
6. **Pool.** The Gaussian renewal process has no refractory floor (1.6% of intervals below 20 ms at 40 pps).
7. **No experimental confrontation yet.** The next validation step is spike-triggered surface MUAPs from decomposed HD-EMG on the same subject against the MRI pipeline, with the normalised error and correlation targets of Lowery et al. [13], Botelho et al. [14].

10 Conclusion

emgforge is a forward model whose every step can be tested, and has been. The synthesis recipe at its centre is the line-source integral done carefully — a monopole-fitted lead field, a second-derivative source, a one-sided tendon window and physical time — and it reproduces first principles exactly. The tests also found the model’s mistakes, and we have listed them rather than smoothed them over. The code, the validation suite and the datasets are public so that both the results and the open items can be checked.

Code and data availability

Code: <https://github.com/dhalatsis/emgforge> (MIT), branch `paper/arxiv-v1`. The validation suite is `scripts/validation/run_all.py`; the figure and dataset scripts are in `paper/figures/`; dataset manifests in `paper/datasets/`.

Author contributions

D.H. designed and implemented the pipeline, the golden method, the activation layer, the validation suite and wrote the report. Noura [surname] developed the harmonic-streamline fibre geometry. Contributions of P. [surname] and D.F. to be stated by the authors.

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